FALL 2024

CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 3 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

By,

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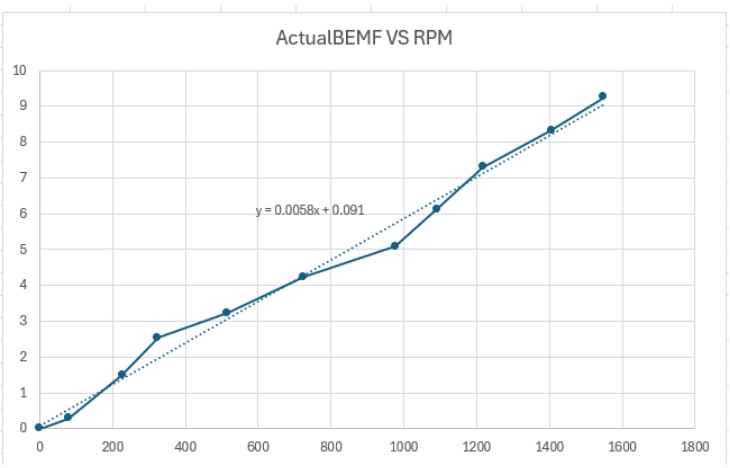
Introduction:

In this lab we will be interfacing a permanent magnet DC motor. The motor has a wheel connected to it that has 32 slits. An IR Led and phototransistor are then implemented so that we can count the number of slits that we passed in order to our current speed or rotations per minute. Additionally, we attempt to further understand the back electromotive force that is produced and how that is affected by increasing and decreasing the PWM. Permanent Magnet DC motors (PMDC motors) are widely used in various applications due to their simplicity, reliability, and ease of control. RPM refers to the number of revolutions a motor's rotor completes in one minute. It is a crucial metric that impacts the motor's performance in several ways. Firstly, RPM directly correlates to the speed at which a motor-driven system operates. For example, a higher RPM in a fan means it will rotate faster and generate more airflow. Applications that require precise speed depend on accurate RPM control for operation. Additionally, rpm influences torque. There is an inverse relationship between RPM and torque. As RPM increases, torque generally decreases and vice versa. Understanding RPM helps optimize motor performance for their intended purpose.

Findings:

The voltage that the motor produces when its rotor revolves in the magnetic field that the permanent magnets have formed is known as the back EMF. Lenz's Law, says that the generated voltage opposes the applied supply voltage because the generated or back EMF always opposes the change that caused it, in this case for our permanent magnet DC motor, the motion of the rotor. The supplied voltage immediately pushes a current through the rotor during startup since the rotor is motionless and the back EMF is zero. High initial torque is the effect of this. When the motor speed increases, the back EMF rises in opposition to the supplied voltage. This lowers the net voltage across the motor. Back EMF nearly matches the applied voltage at the motor's steady-state speed, and the current drawn by the motor is low, just enough to overcome losses like friction and provide torque for the load. At high speeds, back EMF ensures that the motor doesn’t draw excessive current. It improves safety and efficiency by limiting the current level as the motor reaches its operating speed.

PWM (%)	RPM	InverseBEMF	ActualBEMF	BEMFRPM
0	0	1.75	0	0
10	80	1.72	0.29	93
20	228	1.48	1.5	226
30	324	1.35	2.53	388
40	516	1.16	3.23	540
50	726	1.05	4.24	764
60	978	0.86	5.09	980
70	1092	0.677	6.12	1075
80	1218	0.467	7.31	1223
90	1406	0.279	8.318	1440
100	1550	0.13	9.25	1566

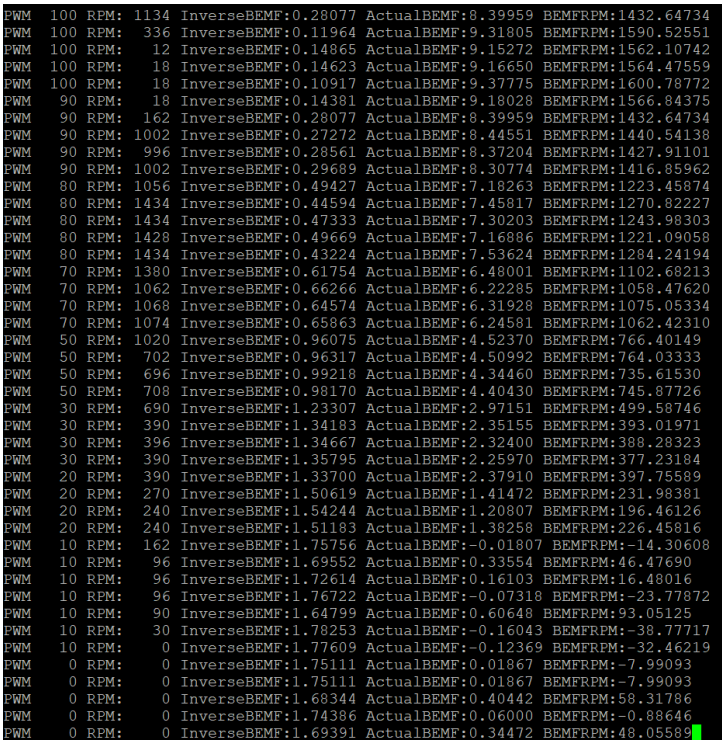


This above table and graph show the values obtained when the motor was operating. The graph is meant to represent that back EMF and RPM are directly proportional. As RPM increases, back EMF rises, and this relationship helps regulate the motor's speed and current draw therefore allow it to safely reach its highest speed without damaging itself.



The yellow line in the picture of the oscilloscope shows the back EMF of the rotor while the purple one shows the collector being activated on the phototransistor.

Results:



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PWM  100 RPM: 1134 InverseBEMF:0.28077 ActualBEMF:8.39959 BEMFRPM:1432.64734
PWM  100 RPM:  336 InverseBEMF:0.11964 ActualBEMF:9.31805 BEMFRPM:1590.52551
PWM  100 RPM:   12 InverseBEMF:0.14865 ActualBEMF:9.15272 BEMFRPM:1562.10742
PWM  100 RPM:   18 InverseBEMF:0.14623 ActualBEMF:9.16650 BEMFRPM:1564.47559
PWM  100 RPM:   18 InverseBEMF:0.10917 ActualBEMF:9.37775 BEMFRPM:1600.78772
PWM   90 RPM:   18 InverseBEMF:0.14381 ActualBEMF:9.18028 BEMFRPM:1566.84375
PWM   90 RPM:  162 InverseBEMF:0.28077 ActualBEMF:8.39959 BEMFRPM:1432.64734
PWM   90 RPM: 1002 InverseBEMF:0.27272 ActualBEMF:8.44551 BEMFRPM:1440.54138
PWM   90 RPM:  996 InverseBEMF:0.28561 ActualBEMF:8.37204 BEMFRPM:1427.91101
PWM   90 RPM: 1002 InverseBEMF:0.29689 ActualBEMF:8.30774 BEMFRPM:1416.85962
PWM   80 RPM: 1056 InverseBEMF:0.49427 ActualBEMF:7.18263 BEMFRPM:1223.45874
PWM   80 RPM: 1434 InverseBEMF:0.44594 ActualBEMF:7.45817 BEMFRPM:1270.82227
PWM   80 RPM: 1434 InverseBEMF:0.47333 ActualBEMF:7.30203 BEMFRPM:1243.98303
PWM   80 RPM: 1428 InverseBEMF:0.49669 ActualBEMF:7.16886 BEMFRPM:1221.09058
PWM   80 RPM: 1434 InverseBEMF:0.43224 ActualBEMF:7.53624 BEMFRPM:1284.24194
PWM   70 RPM: 1380 InverseBEMF:0.61754 ActualBEMF:6.48001 BEMFRPM:1102.68213
PWM   70 RPM: 1062 InverseBEMF:0.66266 ActualBEMF:6.22285 BEMFRPM:1058.47620
PWM   70 RPM: 1068 InverseBEMF:0.64574 ActualBEMF:6.31928 BEMFRPM:1075.05334
PWM   70 RPM: 1074 InverseBEMF:0.65863 ActualBEMF:6.24581 BEMFRPM:1062.42310
PWM   50 RPM: 1020 InverseBEMF:0.96075 ActualBEMF:4.52370 BEMFRPM:766.40149
PWM   50 RPM:  702 InverseBEMF:0.96317 ActualBEMF:4.50992 BEMFRPM:764.03333
PWM   50 RPM:  696 InverseBEMF:0.99218 ActualBEMF:4.34460 BEMFRPM:735.61530
PWM   50 RPM:  708 InverseBEMF:0.98170 ActualBEMF:4.40430 BEMFRPM:745.87726
PWM   30 RPM:  690 InverseBEMF:1.23307 ActualBEMF:2.97151 BEMFRPM:499.58746
PWM   30 RPM:  390 InverseBEMF:1.34183 ActualBEMF:2.35155 BEMFRPM:393.01971
PWM   30 RPM:  396 InverseBEMF:1.34667 ActualBEMF:2.32400 BEMFRPM:388.28323
PWM   30 RPM:  390 InverseBEMF:1.35795 ActualBEMF:2.25970 BEMFRPM:377.23184
PWM   20 RPM:  390 InverseBEMF:1.33700 ActualBEMF:2.37910 BEMFRPM:397.75589
PWM   20 RPM:  270 InverseBEMF:1.50619 ActualBEMF:1.41472 BEMFRPM:231.98381
PWM   20 RPM:  240 InverseBEMF:1.54244 ActualBEMF:1.20807 BEMFRPM:196.46126
PWM   20 RPM:  240 InverseBEMF:1.51183 ActualBEMF:1.38258 BEMFRPM:226.45816
PWM   10 RPM:  162 InverseBEMF:1.75756 ActualBEMF:-0.01807 BEMFRPM:-14.30608
PWM   10 RPM:   96 InverseBEMF:1.69552 ActualBEMF:0.33554 BEMFRPM:46.47690
PWM   10 RPM:   96 InverseBEMF:1.72614 ActualBEMF:0.16103 BEMFRPM:16.48016
PWM   10 RPM:   96 InverseBEMF:1.76722 ActualBEMF:-0.07318 BEMFRPM:-23.77872
PWM   10 RPM:   90 InverseBEMF:1.64799 ActualBEMF:0.60648 BEMFRPM:93.05125
PWM   10 RPM:   30 InverseBEMF:1.78253 ActualBEMF:-0.16043 BEMFRPM:-38.77717
PWM   10 RPM:    0 InverseBEMF:1.77609 ActualBEMF:-0.12369 BEMFRPM:-32.46219
PWM    0 RPM:    0 InverseBEMF:1.75111 ActualBEMF:0.01867 BEMFRPM:-7.99093
PWM    0 RPM:    0 InverseBEMF:1.75111 ActualBEMF:0.01867 BEMFRPM:-7.99093
PWM    0 RPM:    0 InverseBEMF:1.68344 ActualBEMF:0.40442 BEMFRPM:58.31786
PWM    0 RPM:    0 InverseBEMF:1.74386 ActualBEMF:0.06000 BEMFRPM:-0.88646
PWM    0 RPM:    0 InverseBEMF:1.69391 ActualBEMF:0.34472 BEMFRPM:48.05589
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This figure on the left is the output of the terminal showing current PWM percentage, rotations per minute (RPM), BEMF from the microcontroller, actual BEMF given to the Motor, and how BEMF and RPM are related. PMDC motors are vital for their compactness, cost-effectiveness, reliability, and ease of control, making them suitable for a wide range of applications. RPM is a key factor influencing a motor's speed and torque. It helps ensure optimal performance for its intended use. Understanding both PMDC motors and RPM is critical in selecting the right motor for an application and achieving desired performance. Furthermore, back EMF is important because it regulates current flow in permanent magnet DC motors by opposing the applied voltage and limiting motor speed The BEMF RPM as shown is very accurate within the range of the motor speed, displaying that BEMF RPM can be a competitive method to track RPM vs an optical sensor.

Issues:

Some of the issues that came up were when we tried to take the measurements at the poles. For example, at 80% and above, the rpms started getting inaccurate. This is mainly due to us not implementing the Schmitt Trigger. Our triggers on the collector were a right triangle instead of a square wave. If this was implemented, the triggers would have been cleaner and we would have more consistent results at higher speeds. Additionally, at 10% PWM we were not able to get the motor spinning.

Circuit Built:

